Solution to Challenge 6

Invariant Measure for the Logistic Map

(Alligood, Sauer & Yorke, *Chaos: An Introduction to Dynamical Systems*, Ch. 6)

Overview. The tent map T and the logistic map $G(x) = 4x(1-x)$ are conjugate via $C(x) = \frac{1}{2}(1-\cos \pi x)$, i.e. $C \circ T = G \circ C$. The tent map preserves ordinary length (m_1). Because C is a bijection of $[0,1]$, we can *transport* this invariant measure through C to get an invariant measure for G, and then a change of variables turns it into an explicit density $p(x) = \dfrac{1}{\pi\sqrt{x-x^2}}$.

Setup and conventions

Let $T(x) = \begin{cases} 2x, & 0 \le x \le \frac{1}{2} \\ 2(1-x), & \frac{1}{2} \le x \le 1 \end{cases}$ be the tent map T_2 and m_1 ordinary length (Lebesgue measure) on $[0,1]$. As in the book, "measure" means a Borel measure: it is defined on Borel sets (everything built from closed sets by countable unions and complements). The map μ is a *probability measure* if $\mu(\emptyset) = 0$, $\mu([0,1]) = 1$, and $\mu\big(\bigcup_i S_i\big) = \sum_i \mu(S_i)$ for countably many pairwise disjoint sets S_i; it is *f-invariant* if $\mu(f^{-1}(S)) = \mu(S)$ for every Borel set S. Here $f^{-1}(S) = \{x : f(x) \in S\}$ always denotes the preimage (the maps are not invertible).

Properties of C. $C(x) = \frac{1}{2}(1-\cos \pi x) = \sin^2(\pi x/2)$ is continuous with $C'(x) = \frac{\pi}{2}\sin \pi x > 0$ on $(0,1)$, and $C(0) = 0$, $C(1) = 1$. So C is strictly increasing and maps $[0,1]$ *one-to-one onto* $[0,1]$, with a continuous inverse $C^{-1}(y) = \frac{2}{\pi}\arcsin\sqrt{y}$. Consequently C and C^{-1} take Borel sets to Borel sets, and $C^{-1}(S)$ means both the image of S under the inverse function and the preimage of S under C (these coincide).

The conjugacy. (Given in Chapter 3; verified here for completeness.) For $0 \le x \le \frac{1}{2}$,

$$C(T(x)) = \tfrac{1}{2}(1-\cos 2\pi x) = \sin^2 \pi x, \qquad G(C(x)) = 4\sin^2 \tfrac{\pi x}{2}\cos^2 \tfrac{\pi x}{2} = \sin^2 \pi x.$$

For $\frac{1}{2} \le x \le 1$, $C(2-2x) = \frac{1}{2}(1-\cos(2\pi - 2\pi x)) = \frac{1}{2}(1-\cos 2\pi x)$, the same value. So

$$C \circ T = G \circ C \quad \text{on } [0,1]. \tag{1}$$

Invariance of m_1 under T (Exercise T6.14, recalled). For an interval $J = [a,b] \subset [0,1]$,

$$T^{-1}(J) = \left[\tfrac{a}{2}, \tfrac{b}{2}\right] \cup \left[1-\tfrac{b}{2}, 1-\tfrac{a}{2}\right],$$

(the left branch $2x \in [a,b]$ and the right branch $2-2x \in [a,b]$). These two intervals overlap in at most one point, so $m_1(T^{-1}(J)) = \frac{b-a}{2} + \frac{b-a}{2} = b-a = m_1(J)$. Both $S \mapsto m_1(S)$ and $S \mapsto m_1(T^{-1}(S))$ are measures that agree on all intervals, hence agree on all Borel sets:

$$m_1(T^{-1}(S)) = m_1(S) \quad \text{for every Borel set } S. \tag{2}$$

Step 1: $T^{-1}(S) = C^{-1}G^{-1}C(S)$

Let $S \subset [0,1]$. Here $C(S)$ is the image of S, G^{-1} is a preimage, and C^{-1} is the inverse function (equivalently a preimage). We chain equivalences, each justified:

$$\begin{aligned}
x \in T^{-1}(S) &\iff T(x) \in S \\
&\iff C(T(x)) \in C(S) && (C \text{ is one-to-one, so } u \in S \iff C(u) \in C(S)) \\
&\iff G(C(x)) \in C(S) && (\text{by (1)}) \\
&\iff C(x) \in G^{-1}\big(C(S)\big) && (\text{definition of preimage}) \\
&\iff x \in C^{-1}\big(G^{-1}(C(S))\big) && (\text{definition of } C^{-1} \text{ of a set}).
\end{aligned}$$

Hence

$$\boxed{T^{-1}(S) = C^{-1}G^{-1}C(S)}. \tag{3}$$

(The only place we used bijectivity of C is the second line; without it we would only get an inclusion.)

Step 2: $m_1(S) = m_1\big(C^{-1}G^{-1}C(S)\big)$

Let S be a Borel set. Then $C^{-1}G^{-1}C(S)$ is Borel too (C, C^{-1} are homeomorphisms and G is continuous), and by (2) and (3),

$$m_1(S) = m_1\big(T^{-1}(S)\big) = m_1\big(C^{-1}G^{-1}C(S)\big). \qquad \blacksquare$$

Step 3: $\mu(S) = m_1\big(C^{-1}(S)\big)$ is a probability measure

For Borel S, $C^{-1}(S) = \{x : C(x) \in S\}$ is Borel because C is continuous, so $\mu(S)$ is defined, and $0 \le \mu(S) \le 1$ since $C^{-1}(S) \subset [0,1]$.

- $\mu(\emptyset) = m_1(\emptyset) = 0$.
- $\mu([0,1]) = m_1(C^{-1}([0,1])) = m_1([0,1]) = 1$, since C maps $[0,1]$ onto $[0,1]$.
- *Countable additivity.* Let $S_1, S_2, \ldots$ be pairwise disjoint Borel sets. Preimages of disjoint sets are disjoint, so the $C^{-1}(S_i)$ are pairwise disjoint, and $C^{-1}\big(\bigcup_i S_i\big) = \bigcup_i C^{-1}(S_i)$. Therefore

$$\mu\Big(\bigcup_i S_i\Big) = m_1\Big(\bigcup_i C^{-1}(S_i)\Big) = \sum_i m_1\big(C^{-1}(S_i)\big) = \sum_i \mu(S_i),$$

  by countable additivity of m_1.

(Also $\mu(S^c) = m_1\big((C^{-1}S)^c\big) = 1 - \mu(S)$, the complement rule of the book.) So μ is a probability measure on $[0,1]$. $\blacksquare$

Idea: μ is just m_1 pushed through the coordinate change C: the μ-measure of S is the length of the set of "tent-map coordinates" x that C sends into S.

Step 4: μ is G-invariant

Let S be a Borel set. Apply Step 2 to the set $A = C^{-1}(S)$. Since C is a bijection, $C(A) = S$, so Step 2 says

$$m_1\big(C^{-1}(S)\big) = m_1\big(C^{-1}G^{-1}C(C^{-1}(S))\big) = m_1\big(C^{-1}G^{-1}(S)\big).$$

The left side is $\mu(S)$ and the right side is $\mu\big(G^{-1}(S)\big)$ by the definition of μ. Thus

$$\boxed{\mu\big(G^{-1}(S)\big) = \mu(S) \ \text{ for every Borel } S,}$$

i.e. μ is a G-invariant probability measure. ■

Step 5: The density

By definition

$$\mu(S) = m_1\big(C^{-1}(S)\big) = \int_{C^{-1}(S)} 1\,dx.$$

Substitute $y = C(x) = \frac{1}{2}(1 - \cos \pi x)$, so that $x = C^{-1}(y)$ runs over $C^{-1}(S)$ exactly when y runs over S. On $(0,1)$ the map C is a C^1 diffeomorphism and

$$dy = C'(x)\,dx = \tfrac{\pi}{2}\sin(\pi x)\,dx.$$

We must write $\sin \pi x$ in terms of y. From $y = \frac{1}{2}(1 - \cos \pi x)$ we get $\cos \pi x = 1 - 2y$, and since $\sin \pi x \geq 0$ for $x \in [0,1]$,

$$\sin \pi x = \sqrt{1 - (1-2y)^2} = \sqrt{4y - 4y^2} = 2\sqrt{y - y^2}.$$

Hence $dy = \pi\sqrt{y - y^2}\,dx$, that is

$$dx = \frac{dy}{\pi\sqrt{y - y^2}},$$

and therefore

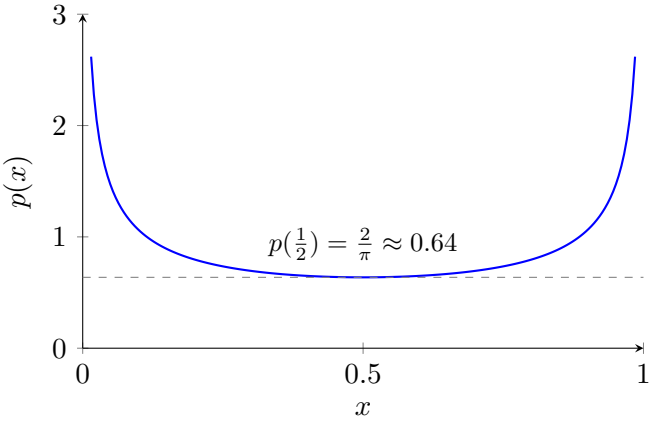
$$\mu(S) = \int_S \frac{1}{\pi\sqrt{y - y^2}}\,dy = \int_S p(y)\,dy, \qquad \boxed{p(y) = \frac{1}{\pi\sqrt{y - y^2}}} \quad (0 < y < 1). \tag{4}$$

The endpoints $y = 0, 1$ do not matter: $\mu(\{0\}) = m_1(C^{-1}\{0\}) = m_1(\{0\}) = 0$, and likewise for 1, and the density is integrable at both ends.

Check that p is a probability density. An antiderivative of p is $\frac{2}{\pi}\arcsin\sqrt{y} = C^{-1}(y)$ (indeed $\frac{d}{dy}\arcsin\sqrt{y} = \frac{1}{2\sqrt{y(1-y)}}$). So for an interval $[\alpha, \beta]$,

$$\int_\alpha^\beta p(y)\,dy = C^{-1}(\beta) - C^{-1}(\alpha) = m_1\big(C^{-1}([\alpha,\beta])\big) = \mu([\alpha,\beta]),$$

which confirms the substitution without any appeal to a general theorem (two measures that agree on intervals agree on all Borel sets). In particular $\int_0^1 p = C^{-1}(1) - C^{-1}(0) = 1$.



The density $p(x) = 1/(\pi\sqrt{x - x^2})$ (Figure 6.15(b)): smallest at $x = \frac{1}{2}$, and tending to ∞ at $x = 0, 1$ (integrably).

An independent check of invariance

For a map with density p, invariance of μ is equivalent to the balance

$$p(y) = \sum_{x \in G^{-1}(y)} \frac{p(x)}{|G'(x)|} \qquad (0 < y < 1),$$

(measure arriving at y equals the measure sitting over its preimages, divided by the local stretching). For $G(x) = 4x(1 - x)$ the two preimages of y are $x_\pm = \frac{1}{2}\big(1 \pm s\big)$ with $s = \sqrt{1 - y}$, and

$$|G'(x_\pm)| = 4|1 - 2x_\pm| = 4s, \qquad x_\pm(1 - x_\pm) = \tfrac{1}{4}(1 - s^2) = \tfrac{y}{4}.$$

Hence $p(x_\pm) = \dfrac{1}{\pi\sqrt{y}/2} = \dfrac{2}{\pi\sqrt{y}}$ and

$$\sum_{\pm} \frac{p(x_\pm)}{|G'(x_\pm)|} = 2 \cdot \frac{2}{\pi\sqrt{y}} \cdot \frac{1}{4\sqrt{1 - y}} = \frac{1}{\pi\sqrt{y(1 - y)}} = p(y),$$

exactly as required. So the density found in Step 5 satisfies the invariance equation directly, independently of the conjugacy argument.

Summary

Step	Result
1	$T^{-1}(S) = C^{-1}G^{-1}C(S)$, from $CT = GC$ and injectivity of C.
2	$m_1(S) = m_1(T^{-1}S) = m_1(C^{-1}G^{-1}C(S))$, since m_1 is T-invariant.
3	$\mu(S) = m_1(C^{-1}S)$ is a probability measure (pullback of m_1 by a bijection C).
4	Apply Step 2 to $C^{-1}(S)$: $\mu(G^{-1}S) = \mu(S)$.
5	$y = C(x)$ gives $dx = dy/(\pi\sqrt{y - y^2})$, so $p(y) = \dfrac{1}{\pi\sqrt{y - y^2}}$.